

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE CODE: MLA0303

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO3:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

Answer:

The reward increases gradually from **10 to 45** over multiple episodes. This indicates that the RL agent is learning from its interaction with the environment.

- Initially, the agent may take poor or random actions, resulting in low rewards.
- As training continues, it updates its policy or value estimates.
- The increasing reward shows that the agent is selecting better actions.
- The policy is therefore improving over time.
- If the reward continues to increase and eventually becomes stable, it suggests convergence toward a good or near-optimal policy.

Conclusion

The trend from 10 → 45 indicates **positive learning performance and policy improvement**. A stable high reward in later episodes would provide stronger evidence that the agent has converged.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. (5 Marks)

Answer:

In the epsilon-greedy strategy:

The agent explores randomly with probability and exploits the best-known action with probability:

Given:

Epsilon Cumulative Reward

0.1 Highest

0.3 Medium

0.5 Lowest

With $\epsilon = 0.5$, the agent explores randomly 50% of the time, so it frequently chooses actions that may not be optimal. Therefore, the cumulative reward is lower.

With $\epsilon = 0.1$, the agent explores only 10% of the time and exploits its learned best action 90% of the time.

Conclusion

$\epsilon = 0.1$ provides the best performance in this experiment because it gives a better balance between exploration and exploitation and achieves the highest cumulative reward.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)

Answer:

Q-Learning consistently obtains slightly higher average rewards than SARSA at **100, 200, and 300 episodes**.

Q-Learning

Q-Learning is an **off-policy** algorithm. It updates its value using the maximum possible future reward:

SARSA

SARSA is an **on-policy** algorithm and learns from the action actually selected:

Since Q-Learning has consistently higher rewards, it is learning a more reward-maximizing policy in the observed environment.

Conclusion

Based on the results:

- **Q-Learning:** Higher average reward
- **SARSA:** Slightly lower average reward
- **Faster convergence:** Q-Learning, based on its better reward performance over the same number of episodes

Thus, **Q-Learning performs better and appears to converge faster** in this experiment.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. (5 Marks)

Answer:

The discount factor determines the importance given to future rewards.

Given values:

A low discount factor such as 0.2 focuses mainly on immediate rewards. A high value such as 0.9 gives greater importance to future rewards.

Since the average reward increases as the discount factor increases, the task benefits from long-term planning.

Conclusion

$\gamma = 0.9$ is the most suitable value because it achieves the highest average reward and helps the agent optimize long-term rewards while still considering immediate rewards.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. (5 Marks)

Answer:

The success rate increases from:

to:

The improvement is:

This significant increase shows that the agent is learning which actions lead to successful outcomes.

Interpretation

- Early training: Policy is still inefficient.
- Later training: Agent makes better decisions.
- At 200 episodes: A 92% success rate indicates strong learning performance.

However, a 92% success rate alone does **not guarantee a perfectly optimal policy**. To confirm convergence, the success rate should remain stable near its maximum over additional episodes.

Conclusion

The agent has learned a **highly effective or near-optimal policy**, but further stable performance is needed to confirm that it is truly optimal.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. (5 Marks)

Answer:

The learning rate controls how much the agent updates its parameters during training.

Given:

Analysis

- **0.01:** Small updates, usually stable but learning may be slow.
- **0.05:** Moderate updates, providing good learning speed and stability.
- **0.1:** Large updates, which may cause oscillation or unstable learning.

Since the agent achieves the highest stability with **0.05**, it provides a good balance between convergence speed and stable training.

Conclusion

Learning rate = 0.05 is the best choice because it provides the best trade-off between:

- Fast convergence
- Stable learning
- Reliable performance

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

(5 Marks)

Answer:

The cumulative reward is highest with **ϵ -greedy exploration**.

1. ϵ -Greedy

The agent:

- Explores randomly with probability
- Exploits the best-known action with probability

This provides a simple balance between exploration and exploitation.

2. Softmax Exploration

Actions are selected based on their estimated values and probabilities. Better actions are more likely to be selected, but weaker actions can still be chosen.

3. Random Exploration

Actions are selected randomly without considering learned action values. This can result in many poor decisions.

Comparison

Strategy	Exploration	Exploitation	Performance
ϵ -Greedy	Controlled	High	Highest reward
Softmax	Probability-based	Moderate	Moderate
Random	Very high	Low	Lower

Conclusion

ϵ -greedy performs best because it effectively balances exploration of new actions with exploitation of the best actions already learned.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. (5 Marks)

Answer:

Initially, the agent requires:

After 300 episodes:

The reduction is:

This means the agent has learned a more efficient path to the goal.

Interpretation

- Initially, the agent explores inefficient paths.
- Through repeated episodes, it learns which actions lead toward the goal.
- The decreasing number of steps indicates improved policy quality.

Conclusion

The reduction from **50 to 15 steps** strongly suggests convergence toward an efficient or near-optimal policy. If the number of steps remains stable around 15 over further episodes, it provides stronger evidence that the agent has converged to the optimal path.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

(5 Marks)

Answer:

The discount factor controls the importance of future rewards.

Interpretation

- $\gamma = 0.4$: Focuses more on immediate rewards.
- $\gamma = 0.7$: Balances immediate and future rewards.
- $\gamma = 0.95$: Gives strong importance to future rewards.

The agent achieves the highest long-term reward with $\gamma = 0.95$. This means that considering future consequences is beneficial in this environment.

The SARSA update is:

A higher γ increases the contribution of future rewards to the update.

Conclusion

$\gamma = 0.95$ is most effective because it allows the agent to optimize long-term rewards rather than making decisions based mainly on immediate rewards.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

(5 Marks)

Answer:

In a stochastic environment, rewards may fluctuate during early episodes because:

- The environment contains randomness.
- The agent is still exploring.
- The policy is changing frequently.
- The agent has limited experience.

After **500 episodes**, the rewards become more stable.

Interpretation

Stable rewards indicate that:

1. The agent has learned a consistent policy.
2. Policy updates are becoming smaller.
3. The agent can handle environmental randomness more effectively.
4. Learning is approaching convergence.

The initial fluctuations are normal because the agent is still exploring and learning from uncertain outcomes.

Conclusion

The stabilization of rewards after 500 episodes indicates **policy improvement, learning convergence, and robustness**. The agent is consistently achieving similar performance despite the stochastic nature of the environment, showing that it has learned a reliable policy.

Code of Conduct:

I Ch. Venkata Lakshmi Hymavathi (192425397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Hymavathi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand